MILITARY SPECIFICATION

TEST SET, SIGHT CURRENT GENERATOR, TS-2660/TSM-100A

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of Sight Current Generator Test Set hereinafter referred to as the "test set" which forms a part of the Electronics System Test Set type AN/TSM-100A, used on the Vulcan Air Defense Weapon System.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODIS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Material: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	-	Parts, Equipment and Tools for Ordnance Material: Packaging of
MIL-C-46357	-	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of

FSC 4931

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center Attn: DRSMC-QA Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:03Z
Check the source to verify that this is the current version before use.

MIL-C-60898		Circuit Card Assemblies, General Specification for
MIL-C-60898/9		Circuit Card Assembly, Sight Current Generator

STANDARDS

MILITARY

MIL-STD-105		Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods

2.1.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

F9333052	-	Panel Assembly, Sight Current Generator Test Unit
F9333067		Schematic Diagram, Electrical, Sight Current Generator, Test Unit

PACKAGING DATA SHEET

9333052	-	Packaging of Test Set, Sight Current Generator

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Fabrication. The test set shall be manufactured in accordance with Drawing F9333052 and drawings pertaining thereto.

3.2 General specifications. Where applicable, the test set shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926

a. Order of precedence

b. Dimensions and tolerances

c. Inorganic protective surface finishes

d. Workmanship

3.2.2 MIL-C-46357

a. Etched circuit assembly. The requirements for etched circuit assembles of MIL-C-46357 shall apply with modification as follows:

(1) Plated etched circuit boards. In addition, assemblies shall not contain copper or nickel visible through hot solder dip, when viewed at 10 times magnification.

(2) Location grid. The component mounting holes shall be within 0.007 inch of the true grid intersection, instead of 0.005.

(3) Transistor and diode leads. Transistor and metal cased diode leads shall be covered with sleeving in areas where there is a possibility of shorting. No color coding of sleeving shall be required.

(4) Drying. The drying process shall not be required.

b. Wiring. The requirements for wiring of MIL-C-46357 shall apply with modification as follows:

(1) Spot ties. Waxed cord may be utilized without the need for varnish on the knots.

(2) Wire termination. Removal of insulation by controlled heat methods shall be acceptable.

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:03Z
Check the source to verify that this is the current version before use.

(3) Chassis cleaning. Compressed air or vacuum may be utilized for cleaning. Vibration shall not be used to clean the chassis.

c. Threaded fasteners. The requirements for threaded fasteners of MIL-C-46357 shall apply except that maximum projection of screws shall be that of the next standard length screw beyond the minimum projection, unless electrical or mechanical interference exists.

d. Marking. The requirements for marking of MIL-C-46357 shall apply except that solvo photo etch process is acceptable.

3.2.3 Circuit card assemblies. The circuit card assemblies shall meet the requirements of MIL-C-60898/9.

3.3 Environmental service conditions.

3.3.1 Vibration (non-operating) . The test set shall be subjected to a simple harmonic motion having excursion as follows:

Frequency (Hz ± 2 Percent)	Excursion (double amplitude) (inches ± 10 percent)
5 to 15	0.06
15 to 25	0.04
25 to 55	0.02

The frequency shall be varied logarithmically between 5 and 55 cycles per second (cps) within 15 minutes and returned to 5 cps in 15 minutes. The motion shall be applied to the test set for a period of 30 minutes in each direction of the 3 mutually perpendicular axis (total of 90 minutes), relative to its normal operational position. Subsequent to vibration, the test set shall not be damaged or adversely affected and shall meet the electrical requirements of 3.4.

3.3.2 Temperatures

3.3.2.1 0perating. The test set shall meet the requirements of 3.4 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.2.2 Storage. The test set shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after-having been exposed to constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 48 hours (after temperature stabilization).

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:03Z
Check the source to verify that this is the current version before use.

3.3.3 Humidity. The test set shall function as specified in 3.4 and shall not suffer any damage or deterioration which shall impair satisfactory performance after exposure to an atmosphere with relative humidity in excess of 95 percent for a period of 10 days.

3.3.4 Shock. The test set shall function as specified in 3.4 and shall show no evidence of damage or be adversely affected after being subjected to a total of 18 shocks consisting of three shocks in opposing directions along each of the three mutually perpendicular axis. Each shock shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 ± 1 millisecond.

3.3.5 Altitude

3.3.5.1 Operating. The test set shall not be damaged or adversely affected when exposed to conditions encountered at an altitude of 10,000 feet above sea level (20.6 inches Hg). During and subsequent to such exposure the test set shall meet the requirements of 3.4.

3.3.5.2 Non-operating. The test set shall meet the requirements of 3.4 after exposure to conditions encountered at 40,000 feet above sea level (5.5 inches Hg) for a period of not less than 10 minutes. The test set. shall not be damaged or be adversely affected due to such exposure.

3.4 Electrical.

3.4.1 Electrical continuity. Each conductor shall be electritally continuous as specified on the schematic Drawing 9333067. The symbols referenced in subsequent paragraphs pertain to this schematic and are used therein

3.4.2 Conditions of operation. Unless otherwise specified, the following requirements shall be met met with the conditions listed below:

a. Energizing voltages

(1) A voltage of +22,0 ± 0.1 volts, direct-current (Vdc), applied between pins F (+) and E (-) of connector J103C

(2) A voltage of -22.00 ± 0.15 Vdc applied between pins G (-) and K (+) of connector J102.

(3) A voltage of +10.0 ± 0.1 Vdc applied between pins F (+) and C (-) of connector J102.

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:03Z
Check the source to verify that this is the currentversion before use.

(4) A voltage of -10.0 ± 0.1 Vdc applied between pins E (-) and C (+) of connector J102.

b. Control settings

(1) Mode (S11) - radar ready

(2) System power (S12) - on

(3) Circuit breaker (CB1) - closed

(4) Range/Range Rate (S15) - manual

3.4.3 RADAR RANGE sianal. There shall be an analog signal continuously variable from 0.0 ± 0.1 Vdc to 10.0 ± 0.1 Vac at J2-W(+) to J2-V as the RADAR RANGE knob is varied from 0.00 to 10.00.

3.4.4 RADAR RANGE RATE signal. With the INBOUND/OUTBOUND switch in the "INBOUND" position, there shall be analog signal continuously variable from 0.0 ± 0.1 VDC to -10.0 ± -.1 Vdc at K2-K(-) to J2-V as the RADAR RANGE RATE knob is varied from 0.00 to 10.00.

3.405 SYSTEM POWER. When the SYSTEM POWER switch is in the "ON" position, the SYSTEM POWER indicator shall LIGHT and the resistance shall be less than 2.5 OHMS between the connector pins below. When the switch is "OFF", the SYSTEM POWER indicator shall be EXTINGUISHF and the resistance shall be greater than 500 KILOHMS between the following pins:

J103-C to J7-C
J103-G to J7-C
J103-B to J5-A
J103-H to J5-H
J103-K to J5-G

3.4.6 MODE. The voltages per Table I shall be present at the specified connector pins only when the MODE switch is in the specified position. With the RANGE/RANGE RATE switch in the MANUAL position and the MODE switch in the RADAR READY position, the resistance between the connector pins shown below shall be less than 2.5 OHMS. When the MODE switch is set to K25 EXTERNAL position, the resistance between the connector pins shown below shall be greater than 500 KILOHMS.

J102-D to J2-X
J102-F to J2-G
J102-E to J2-H
J102-G to TP-21

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

TABLE I. Mode test

MODE Switch Position	J7 Connector Pins (J7-R Return)						
	E	a	b	d	e	m	r
GROUND	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RADAR READY	0.0	0.0	22.0	0.0	22.0	0.0	22.0
K26 MANUAL	22.0	22.0	0.0	0.0	22.0	0.0	22.0
K25 EXTERNAL	0.0	0.0	0.0	0.0	0.0	22.0	22.0
TEST	0.0	0.0	22.0	22.0	22.0	0.0	22.0

3.4.7 ACTION SWITCH. With the MODE switch set to the GROUND position, 22.0 ± 0.1 vdc all be present between J7-A(+) and J7-R and 0.0 ± 0.1 Vdc shall be present between J7-k(+) and J7-R when the ACTION SWITCH is in the "ON" position. With the MODE switch set to the RADAR READY position, 22.0 ± 0.1 Vdc shall be present between J7-A(+) and J7-R and between J7-k(+) and J7-R when the ACTION SWITCH is in the "ON" position,

3.4.8 LOCKON. With the LOCKON switch in the "ON" position, the resistance J2-C and J2-B shall be greater than 500 KILOHMS and with the switch in the "OFF" position the resistance between J2-C and J2-B shall be less than 2.5 OHMS.

3.4.9 ECM. With the ECM pushbutton switch DEPRESSED, the resistance between J2-P and J2-R shall be less than 2.5 OHMS and with the ECM pushbutton switch RELEASED, the resistance between J2-P and J2-R shall be greater than 500 KILOHMS

3.4.10 SIGHT CAGE BUTTON. With the SIGHT CAGE BUTTON pushbutton switch DEPRESSED, the resistance between J7-c and J7-s shall be less than 2.5 OHMS and with the pushbutton switch RELEASED, the resistance between J7-c and J7-s shall be greater than 500 KILOHMS.

3.4.11 FOOT SWITCH. With the FOOT SWITCH IN THE "ON" position, the resistance between J3-A and J3-13 shall be less than 2.5 OHMS and with the switch in the "OFF" position, the resistance between J3-A and J3-B shall be greater than 500 KILOHMS.

3.4.12 ELEVATION GREATER THAN 80°. With the ELEVATION GREATER THAN 800 pushbutton switch DEPRESSED, the resistance between J7-K and J7-R shall be greater than 500 KILOHMS and with the pushbutton switch RELEASED, tke resistance between J7-K and J7-R shall be less than 2.5 OHMS.

3.4.13 EXTERNAL READY TO FIRE. With the EXTERNAL READY TO FIRE pushbutton switch DEPRESSED, the resistance between J7-g and

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

J7-R shall be less than 2.5 OHMS and when the switch is RELEASED, the resistance between J7-g and J7-R shall be greater than 500 KILOHMS .

3.4.14 LEAD ANGLE GREATER THAN 25°. With the MODE switch set to the RADAR READY position and the LEAD ANGLE GREATER THAN 25° pushbutton switch DEPRESSED, -8.42 ± .15 Vdc shall be present between J4-L(-) and J7-P and between J4-V(-) and J7-P, With the pushbutton switch RELEASED, there shall be an OPEN CIRCUIT between J4-L and J7-P and between J4-V and J7-P,

3.4.15 RANGE GREATER THAN 3000M. With a voltage of +28.0 ± 0.1 Vdc applied between pins F (+) and E (-) of connector J103, +15.4 ± 1.0 Vdc shall be present between J2-W(+) and J7-P with the RANGE GREATER THAN 3000M pushbutton switch DEPRESSED.

3.4.16 INBOUND-OUTBOUND. With RADAR RANGE knob set at 10.00 and the INBOUND-OUTBOUND switch in the OUTBOUND position +10.0 ± 0.1 Vdc shall be present between J2-K(+) and J2-V.

3.4.17 READY TO FIRE indicator. With 22.0 ± 0.1 Vdc applied between J6-L(+) and J6-N, the READY TO FIRE indicatcr shall LIGHT.

3.4.18 ANTENNA SERVO COMMAND indicator. With 22.0 ± 0.1 Vdc applied between J2-Y(+) and J2-U, the ANTENNA SERVO COMMAND indicator shall LIGHT.

3.4.19 TEST indicator. With 22.0 ± 0.1 Vdc applied between J7-V(+) and J7-W, the TEST indicator shall LIGHT.

3.4.20 K23 10 SEC TD indicator. With 22.0 ± 0.1 Vdc applied between J7-p(+) and J7-R, the K23 10 SEC TD indicator shall LIGHT.

3.4.21 RADAR READY indicator. With 22.0 ± 0.1 Vdc applied between J7-X(+) and J7-R, the RADAR READY indicator shall LIGHT.

3.4.22 RADIATE COMMAND indicator. With 22.0 ± 0.1 Vdc applied between J2-Z(+) and J7-R, the RADIATE COMMAND indicator shall light.

3.4.23 K26 MANUAL indicator. The K26 MANUAL indicator shall LIGHT only when the MODE switch is in the K26 MANUAL position.

3.4.24 K22 ACTION SWITCH indicator. With 22.0 ± 0.1 Vdc applied between J2-Y(+) and J7-R(-), the K22 ACTION SWITCH indicator shall LIGHT . It shall also LIGHT when +22.0 ± 0.1 Vdc is applied between J7-j(+) and J7-R(-).

3.4.25 K25 EXTERNAL indicator. With 22.0 ± 0.1 Vdc applied between J7-j(+) and J7-h, the K25 EXTERNAL indicator shall LIGHT.

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:03Z
Check the source to verify that this is the current version before use.

3.4.26 K27 READY TO FIRE indicator. With +22.0 ± 0.1 Vdc applied between J1-e (+) and J7-R, the K27 READY TO FIRE indicator shall not LIGHT . With +22.0 ±0.1 Vdc applied between J1-e(+) and J7-R, and between J1-b(+) and J7-R the indicator shall LIGHT. The indicator shall remain LIT when +22.0 ± 0.1 Vdc is removed from J1-b and shall remain LIT until 22.0 ± 0.1 Vdc is removed from J1-e.

3.4.27 SIGHT CAGE indicator. With the MODE switch set to the RADAR READY position and -1 .25 ± 0.01 Vdc applied between J6-D(-) and J7-R, the SIGHT CAGE indicator shall not LIGHT. With -1.55 ± 0.01 Vdc applied between J6-D(-) and J7-R, the indicator shall LIGHT.

3.4.28 Torque Motor. A series inductance and resistance of 0.15 henry ±20% at 400 Hz and 37 ohms ±15%, respectively, shall exist between J6-K and J6-T. The resistance between J6-K and J6-B shall be 2.0 ± 001 OHMS.

3.4.29 Magnet. With a voltage source, set at 20.0 ± 0.1 Vdc connected between J6-M(+) and J6-A(-) , there shall be +1.88 ± 0.35 Vdc between J6-P(+) and J6-D(-). With a current source set at 1 ± 0.1 ampere, connected between J6-A(+) and J6-M(-) there shall be +0.9 ± 0.5 Vdc between J6-A(+) and J6-M(-).

3.4.30 MANUAL RANGE signal. With voltage and load applied per Table II, the following voltages shall be present between J7-S(+) and J7-U for the indicated MANUAL RANGE knob settings.

SETTING	VOLTAGE
500	+2.50 ± 0.05 Vdc
1500	+7.50 ± 0.05 Vdc
2500	+12.50 ± 0.05 Vdc

3.4.31 MANUAL RANGE RATE signal. With voltage and load applied per Table II, the following voltages shall be present between J7-T(+) and J7-U for the indicated MANUAL RANGE RATE knob,

SETTING	VOLTAGE
0	+7.50 ± 0.05 Vdc
300	+3.75 ± 0.05 Vdc
600	0.00 ± 0.05 Vdc

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:03Z
Check the source to verify that this is the current version before use.

TABLE II. Manual range rate signal test

SECTION	APPLIED VOLTAGE	APPLIED LOADS
3.4.30	15.00 ± 0.05 Vdc between J7-f(+) and J7-U(-)	none
3.4.31	Same as 3.4.30	none
3.4.32	10.00 ± 0.05 Vdc between J4-A(+) and J4-R(-)	7 K-ohms between J4-B and J4-R
3.4.33	10.00 ± 0.05 Vdc between J4-K(+) and J4-M(-) and J4-Z(-)	50 K-ohms between J4-L and J4-K
3.4.34	10.00 ± 0.05 Vdc between J4-c(+) and J4-J(-) and J4-$\overline{E}$(-) and J4-F(-) and J4-Y(-) 10.00 ± 0.05 Vdc between J4-W(+) and J4-G(+) J4-c(-)	50 K-ohms connected across DVM

3.4.32 COSINE signal. With voltage and load applied per Table II, the following voltages shall be present between J4-B(+) and J4-R for the indicated COSINE knob settings.

SETTING	VOLTAGE
0	10.00 ± .05 Vdc
60	5.00 ± .05 Vdc

3.4.33 LEAD ANGLE TRAVERSE signal. With the voltage and load applied per Table II, the following voltages shall be present between J4-L(+) and J4-K for the indicated LEAD ANGLE TRAVERSE knob settings.

SETTING	VOLTAGE
15	-3.600 ± 0.035 Vdc
0	0.0 + 0.1 Vdc
-15	-3.600 ± 0.035 Vdc

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:03Z
Check the source to verify that this is the current version before use.

3.4.34 LEAD ANGLE EL signals. With the voltages and load applied per Table II, the following voltages shall be present between J4-v(+) and J4-c, J4-H(+) and J4-c and J4-X(+) and J4-c for the indicated LEAD ANGLE EL knob settings.

SETTING	VOLTAGE (J4-V to J4-c)	VOLTAGE (J4-H to J4-C)	VOLTAGE (J4-X to J4-c)
15	-3.600 ± 0.035 Vdc	+6.00 ± 0.15 Vdc	+2.16 ± 0.21 Vdc
0	0.0 ± 0.1 Vdc	0.0 ± .065 Vdc	0.0 ± 0.1 vdc
-15	-3.600 ± 0.035 Vdc	-6.00 ± 0.15 Vdc	-2.16 ± 0.21 Vdc

3.4.35 5.0 Amperes circuit breaker. The circuit breaker shall not open within a 2 minute period when subjected to a current of 5.75 ± 0.05 amperes DC. With a 200 percent overload of the circuit breaker rated current (10.0 ± 0.1 amperes) , the circuit breaker shall open within 15 to 55 seconds.

3.4.36 K3 Relay.

3.4.36.1 K3 Energized. With a voltage of +22.0 ± 0.1 Vdc applied between pins F (+) and E (-) of connector J103, the SYSTEM POWER switch in the "ON" position, and the RANGE/RANGE RATE switch set to the *'AUTO" position; the resistance shall be less than 2.5 OHMS between the connector pins shown below.

J10-K to J2-K
J10-H to J2-X
J10-F to J2-G
J10-J to J2-W
J10-G to J2-H

A voltage of +22.0 ± 0.1 Vdc, shall be present between J10-B(+) and J10-E(-).

3.4.36.2 K3 Inactive. With the RANGE/RANGE RATE switch in the "MANUAL" position, the resistance shall be greater than 500 KILOHMS between the following connector pins.

J10-K to J2-K
J10-J to J2-W
J10-H to J2-X
J10-G to J2-H
J10-F to J2-G

An OPEN CIRCUIT shall be present between J10-B(+) and J10-E(-).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

3.4.37 Electrical performance for temperature extremes. The following requirements shall apply for the conditions of 3.3.2 (as applicable).

3.4.37.1 LEAD ANGLE GREATER THAN 25°. With the MODE switch set to the RADAR READY position and the LEAD ANGLE GREATER THAN 25° push-button switch DEPRESSED, the voltage present between J4-L(-) and J7-P shall be -8.42 ± 0.50 Vdc.

3.4.37.2 RANGE GREATER THAN 3000M. With a voltage of 28.0 ± 0.1 Vdc applied between pins F (+) and E (-) of connector J103, and the RANGE GREATER THAN 3000M pushbutton switch DEPRESSED, the voltage present between J2-W(+) and J7-P shall be +15.4 ± 1.5 Vdc.

3.4.37.3 SIGHT CAGE indicator. With -1.10 ± 0.01 Vdc applied between J6-D(-) and J7-R, the SIGHT CAGE indicator shall NOT LIGHT. With -1.70 ± 0.01 Vdc applied between J6-D(-) and J7-R, the indicator shall LIGHT.

3.5 First Article. Unless otherwise specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.6 Workmanship. The requirements of MIL-STD-454 and other applicable specifications shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Unless otherwise specified in the detail specification, or contract, the following types of inspection shall be conducted on the unit or product:

a. First article inspection (4.3)

b. Quality conformance inspection (4.4)

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article as designated by the contracting officer (see 6.2) for evaluation

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

in accordance with the provisions of 4.3.2. The first article sample shall consist of 1 test set.

4.3.2 Inspection to be performed. First article sample may be subjected by the Government to any or all of the examinations and tests specified in 4.4.2 and 4,4.4 of this specification and to any or all requirements of the applicable drawings.

4,3.3 Rejection. In any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component, or test specimen in the sample to comply with any of the stated requirements.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. The term “inspection lot” is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests. Except where otherwise specified herein, components and subassemblies shall be inspected in accordance with the “Inspection provisions for components and subassemblies” and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables XII and IV and methods of inspection in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. Workmanship shall be in accordance with MIL-STD-454 and other applicable specifications.

4.4.2.1 Components and subassemblies. Printed circuit subassemblies shall be inspected in accordance with the inspection provisions of MIL-C-60898. All *other* components and subassemblies shall be inspected to the applicable requirements of MIL-F-13926.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

4.4.2.2 Control tests. One test set shall be selected at random as a control sample from each 100 produced. The control samples shall meet the requirements and tests of Table III after they have met the requirements and tests of Table IV.

TABLE III. Control test

Test	Requirement	Test Procedures
Fabrication	3.1	Applicable drawings - visual
General Specifications	3.2	MIL-F-13926/MIL-C-46357 - visual
Vibration	3.3.1	4.5.1.1, 4.5.1.2
Temperature	3.3.2, 3.4.37	4.5.1.1, 4.5.1.3
Humidity	3.3.3	4.5.1.1, 4.5.1.4
Shock	3.3.4	4.5.1.1, 4.5.1.5
Altitude	3.3.5	4.5.1.1, 4.5.1.6

4.4.3 Rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4.4 Functional tests. The requirements and tests in Table IV shall be inspected on a 100% basis. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a. Temperature. +50° to +90° F -
b. Relative humidity. 90% RH max.
c. Pressure. 28 to 32 inches of Hg

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

TABLE IV. Functional test
(100 percent inspection)

Characteristic	Requirement	Procedure
Continuity test	3.4.1	4.5.2.1
Conditions of operation	3.4.2	4.5.2.2.1
RADAR RANGE signal	3.4.3	4.5.2.2.2
RADAR RANGE RATE signal	3.4.4	4.5.2.2.3
SYSTEM POWER	3.4.5	4.5.2.2.4
MODE	3.4.6	4.5.2.2.5
ACTION	3.4.7	4.5.2.2.6
LOCKON	3.4.8	4.5.2.2.7
ECM	3.4.9	4.5.2.2.8
SIGHT CAGE BUTTON	3.4.10	4.5.2.2.9
FOOT SWITCH	3.4.11	4.5.2.2.10
ELEVATION GREATER THAN 80°	3.4.12	4.5.2.2.11
EXTERNAL READY TO FIRE	3.4.13	4.5.2.2.12
LEAD ANGLE GREATER THAN 25°	3.4.14	4.5.2.2.13
RANGE GREATER THAN 3000M	3.4.15	4.5.2.2.14
INBOUND-OUTBOUND	3.4.16	4.5.2.2.15
READY TO FIRE	3.4.17	4.5.2.2.16
ANTENNA SERVO COMMAND	3.4.18	4.5.2.2.17
TEST	3.4.19	4.5.2.2.18
K23 10 SEC TD	3.4.20	4.5.2.2.19
RADAR READY	3.4.21	4.5.2.2.20
RADIATE COMMAND	3.4.22	4.5.2.2.21

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:03Z
Check the source to verify that this is the current version before use.

TABLE IV. Functional test - continued
(100 percent inspection)

Characteristic	Requirement	Procedure
K26 MANUAL	3.4.23	4.5.2.2.22
K22 ACTION SWITCH	3.4.24	4.5.2.2.23
K25 EXTERNAL	3.4.25	4.5.2.2.24
K27 READY TO FIRE	3.4.26	4.5.2.2.25
SIGHT CAGE	3.4.27	4.5.2.2.26
Torque Motor	3.4.28	4.5.2.2.27
Magnet	3.4.29	4.5.2.2.28
MANUAL RANGE signal	3.4.30	4.5.2.2.29
MANUAL RANGE RATE signal	3.4.31	4.5.2.2.30
COSINE signal	3.4.32	4.5.2.2.31
LEAD ANGLE TRANSVERSE signal	3.4.33	4.5.2.2.32
LEAD ANGLE EL signal	3.4.34	4.5.2.2.33
CIRCUIT BREAKER	3.4.35	4.5.2.2.34
K3 RELAY	3.4.36	4.5.2.2.35

4.4.5 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the referenced specifications and standards cited in 4.4.2 and 4.4.4. Inspection equipment shall be as listed in Table V. The contractor shall submit inspection equipment designs for approval in accordance with the term of the contract. See 6.3 of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

TABLE V. Inspection equipment list.

Item		Description
1.	MULTIMETER	Ohmmeter function: Rx1 to RX10,000 scales with accuracy and resolution to 5% or better. Operated from: 1.5 Vdc or greater with an internal impedance of less than 12 ohm for the RX1 scale, and from: 6 Vdc or greater with an internal impedance of less than 200 K ohms on the RX10,000 scale. D.C. voltmeter function: 2.5 volt to 50 volt scales with 2% accuracy and 20,000 ohms/volt typical characteristic.
2.	DIGITAL VOLTMETER (DVM)	A range of at least .0010 to 999.9 volts dc, or four significant figures. Accuracy of at least ±0.2 percent of reading. Input impedance of at least 10 megohms on all ranges.
3.	IMPEDANCE BRIDGE	DC and AC operation; 100 micro-henry to 5 henry at 1% accuracy and resolution; 10 ohm to 100 K ohms at 1% accuracy and resolution.
4.	MEGGER	500 Vdc ± 10%, O to 100 megohms with accuracy of ± 10%.
5.	RESISTOR	Decade resistor, O to 99 K ohms or greater, adjustable to ± 1 K ohm with 0.1% or better accuracy.
6.	POWER SUPPLY #1	Nominal value of plus 24 Vdc; 0-30 Vdc. adjustable voltage. Current rating 1.5 ampere. Less than 0.5% ripple.
7.	POWER SUPPLY #2	Nominal value of plus 10 Vdc; 0-30 Vdc, adjustable voltage. Current rating 500 M.A. Less than 0.1% ripple.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

TABLE V. Inspection equipment list - continued.

Item	Description
8. POWER SUPPLY #3	Nominal value of minus 10 Vdc; 0-30 Vdc, adjustable voltage. Current rating 500 M.A. Less than 0.1% ripple.
9. POWER SUPPLY #4	Nominal value of minus 23 Vdc; 0-30 Vdc, adjustable voltage. Current rating 500 M.A. Less than 0.1% ripple.
10. POWER SUPPLY #5	A D.C. power supply and an ammeter indicating load current and an adjustable current limiting control. The capacity will be 75 amperes or greater and O to 50 Vdc.

4.5 Methods of inspection.

4.5.1 Environmental.

4.5.1.1 Test conditions. The test set shall be mounted in an upright position in a manner similar to that in which it will be in service and shall be subjected to the environmental tests of the following subparagraphs. The atmospheric pressure shall be room ambient. Except for the high and low temperature tests or the humidity tests, as applicable, the temperature and humidity shall be room ambient.

4.5.1.2 Vibration. The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus ox minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1. Subsequent to the vibration test, the test set shall show no damage or defect resulting from the test when examined by visual and tactile means. The test set shall then be tested to determine compliance with requirements specified under 3.3.1.

4.5.1.3 Temperature. Test in accordance with Procedures I and II of Method 501 in MIL-STD-810 for high temperature operating and non-operating under conditions specified in 3.3.2. Test in accordance with Procedures I and II of Method 502, in MIL-STD-810 for low tempera-

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

ture operating and non-operating under conditions specified in 3.3.2, and verify requirements of 3.3.2.1 and 3.3.2.2.

4.5.1.4 Humidity. The humidity test shall be performed in accordance with Procedure III of Method 507 in MIL-STD-810, except that in Step 7 the operational checkout shall be performed at the end of the tenth cycle only and at standard ambient conditions.

4.5.1.5 Shock. Test in accordance with Procedure I Method 516 of MIL-STD-810 except that shock pulses shall be as specified in 3.3.4. After the shock impulses the test set shall be tested as specified in 4.5.2 for compliance with 3.3.4.

4.5.1.6 Altitude (operating and non-operating). The altitude test shall be performed in accordance with Procedures II and III of Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.5.2 for compliance with 3.3.5.

4.5.2 Electrical tests

4.5.2.1 Continuity test. Continunity shall be inspected by ohmmeter test, using item 1 of 4.4.5, and visual examinations to verify conformance to 3.4.1.

4.5.2.2 Functional tests

4.5.2.2.1 Conditions of operation. Unless otherwise specified, all subsequent tests will be performed with the D.C. power supply connections and control settings as follows: Set POWER SUPPLY #1, item 6 of 4.4.5, to +22.0 ± 0.1 Vdc and connect to J103F(+) and J103E(-). Set POWER SUPPLY #2, item 7 of 4.4.5, to +10.0 ± 0.1 Vdc and connect to J102F(+) and J102C(-). Set POWER SUPPLY #3, item 8 of 4.4.5, to +10.0 ± 0.1 Vdc and connect to J102K(-) and J102C(+). Set POWER SUPPLY #4, item 9 of 4.4.5, to +22.0 ± 0.15 Vdc and connect to J102G(-) and J102K(+). Set the MODE control to "RADAR READY". Set the SYSTEM POWER switch to "ON". Close the circuit breaker. Set the RANGE/RANGE RATE switch to "MANUAL"

4.5.2.2.2 RADAR RANGE signal. Connect DVM, item 2 of 4.4.5, between J2-W(+) and J2-V. Vary the RADAR RANGE from 0.00 to 10.00 and verify 3.4.3.

4.5.2.2.3 RADAR RANGE RATE signal. Connect the DVM, item 2 of 4.4.5 between J2-K(+) and J2-V. Place the INBOUND/OUTBOUND switch in the “INBOUND” position. Vary the RADAR RANGE RATE from 0.00 to 10.00 and verify 3.4.4.

4.5.2.2.4 SYSTEM POWER. With the SYSTEM POWER switch in the “ON” position and with the MULTIMETER, item 1 of 4.4.5, connected between the pins specified in 3.4.5, verify the SYSTEM POWER

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

indicator is LIT, and verify 2.5 OHMS, or less, between the pins as specified in 3.4.5. With the SYSTEM POWER switch in the "OFF" position, verify that the SYSTEM POWER indicator is EXTINGUISHED, and verify 500 K-OHMS, or more, between the pins as specified in 3.4.5 Set SYSTEM POWER switch to the “ON” position.

4.5.2.2.5 MODE switch. With the DVM, item 2 of 4.4.5, verify the voltages specified in Table I for each MODE switch position. With the RANGE/RANGE RATE switch in the MANUAL position, and the MODE switch in the RADAR READY position, use the MULTIMETER, item 1 of 4.4.5, and verify that the resistance between conector pins specified in 3.4.6 are less than 2.5 OHMS. With the MODE switch set to K25 EXTERNAL position, verify 500 K-OHMS, or more, between the pins as specified in 3.4.6.

4.5.2.2.6 ACTION switch. With the DVM, item 2 of 4.4.5, connected between J7-A(+) and J7-R, verify +22.0 ± 0.1 Vdc when the ACTION SWITCH is in the "ON" position, and the MODE switch is in both the GROUND and RADAR READY positions. Transfer the DVM PLUS LEAD to J7-k. With the ACTION SWITCH in the “ON” position, verify 22.0 ± 0.1 Vdc is present when the MODE switch is in the RADAR READY position, and 0.0 ± 0.1 Vdc when the MODE switch is in the GROUND position.

4.5.2.2.7 LOCKON switch. With the MULTIMETER, item 1 of 4.4.5, connected between J2-C and J2-B, verify greater than 500 K-OHMS with the LOCKON switch in the “ON” position and less than 2.5 OHMS with the LOCKON switch in the “OFF” position.

4.5.2.2.8 ECM pushbutton switch. With the MULTIMETER, item 1 of 4.4.5, connected between J2-P and J2-R, verify less than 2.5 OHMS when ECM pushbutton switch is DEPRESSED, and Greater than 500 K-OHMS when the ECM pushbutton switch is RELEASED.

4.5.2.2.9 SIGHT CAGE BUTTON pushbutton switch. With the MULTIMETER, item 1 of 4 4 5, connected between J7-c and J7-s, verify less than 2.5 OHMS with the SIGHT CAGE BUTTON-pushbutton switch DEPRESSED and greater than 500 K-OHMS with the switch RELEASED.

4.5.2.2.10 FOOT SWITCH switch. With the MULTIMETER, item 1 of 4,4.5, connected between J3-A and J3-B, verify less than 2.5 OHMS with the FOOT SWITCH switch in the “ON” position, and greater than 500 K-OHMS with the switch in the “OFF” position.

4.5.2.2.11 ELEVATION GREATER THAN 80° pushbutton switch. With the MULTIMETER, item 1 of 4 4 5, connected between J1-K and J7-R, verify greater than 500 K-OHMS with the ELEVATION GREATER THAN 80° pushbutton switch DEPRESSED and less than 2.5 OHMS with the switch RELEASED.

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:03Z
Check the source to verify that this is the current version before use.

4.5.2.2.12 EXTERNAL READY TO FIRE pushbutton switch. With the MULTIMETER, item 1 of 4.4.5, connected between J7-g and J7-R, verify less than 2.5 OHMS with the EXTERNAL READY TO FIRE pushbutton switch DEPRESSED and greater than 500 K-OHMS with the switch RELEASED.

4.5.2.2.13 LEAD ANGLE GREATER THAN 25° pushbutton switch. With the DVM. item 2 of 4.4 5, connected between J4-L(+) and J7-P, and the MODE switch set to the RADAR READY position, verify -8.42 ± 0.15 Vdc with the LEAD ANGLE GREATER THAN 25° pushbutton switch DEPRESSED and verify an OPEN CIRCUIT with the switch RELEASED. Transfer the DVM PLUS LEAD to J4-V. Verify -8.42 ± 0.15 Vdc with the swith DEPRESSED and an OPEN CIRCUIT with the switch RELEASED.

4.5.2.2.14 RANGE GREATER THAN 3000M pushbutton switch. With POWER SUPPLY #l, item 6 of 4.4.5, apply +28 .0 2 0.1 Vdc between pins F (+) and E (-) of connector J103. with the DVM, item 2 of 4.4.5, connected between J2-W(+) and J7-P, verify 15.4 ± 1.0 Vdc with the RANGE GREATER THAN 3000M pushbutton switch DEPRESSED. Return POWER SUPPLY #1, item 6 of 4.4.5, to +22.0 ± 0.1 Vdc and connect to J103-F (+) and J103-E (-).

4.5.2.2.15 INBOUND/OUTBOUND switch. With the DVM, item 2 of 4.4.5, connected between J2-K(+) and J2-V, the RADAR RANGE knob turned to 10.0, and the INBOUND\OUTBOUND switch to OUTBOUND, verify +10.0 ± 0.1 VdC between the pins.

4.5.2.2.16 READY TO FIRE indicator. With POWER SUPPLY #2, item 7 of 4.4.5, apply 22.0 ± 0.1 Vdc between pins J6-L(+) and J6-N, verify that the READY TO FIRE indicator is “LIT”.

4.5.2.2.17 ANTENNA SERVO COMMAND indicator. With POWER SUPPLY #2, item 7 of 4. 4.5, apply 22.0 ± 0.1 Vdc between pins J2-Y(+) and J2-U, verify that the ANTENNA SERVO COMMAND indicator is “LIT”.

4.5.2.2.18 TEST indicator, With POWER SUPPLY #2, item 7 of 4.4.5, apply 22.0 ± 0.1 Vdc between pins J7-V(+) and J7-W, verify that the TEST indicator is “LIT”.

4.5.2.2.19 K23 10 SEC TD indicator. With POWER SUPPLY #2, item 7 of 4.4.5, apply 22 0 ± 0 1 Vdc between pins J7-p(+) and J7-R, verify that the K23 10 SEC TD indicator is "LIT"

4.5.2.2.20 RADAR READY indicator. With POWER SUPPLY #2, item 7 of 4.4.5, apply 22.0 ± O.1 Vdc between pins J7-X(+) and J7-R, verify that the RADAR READY indicator is “LIT”.

4.5.2.2.21 RADIATE COMMAND indicator. With POWER SUPPLY #2, item 7 of 4.4.5, apply 22.0 ± 0.1 Vdc between pins J2-Z(+) and J7-R, verify that the RADIATE COMMAND indicator is "LIT"

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

4.5.2.2.22 K26 MANUAL indicator. Rotate the MODE switch through all positions. Verify that the K76 MANUAL indicator is "LIT" Only when the MODE switch is in the K26 MANUAL position.

4.5.2.2.23 K22 ACTION SWITCH indicator. With POWER SUPPLY #2, item 7 of 4.4.5, apply 22.0 ± 0.1 VDC between pins J2-Y(+) and J7-R(-) , verify that the K22 ACTION SWITCH indicator is “LIT”, With POWER SUPPLY #2 apply 22.0 ± 0.1 Vdc between pins J7-j(+) and J7-R, verify that the indicator is again “LIT”.

4.5.2.2.24 K25 EXTERNAL indicator. With POWER SUPPLY #2, item 7 of 4.4.5, apply 22.0 ± 0.1 Vdc between pins J7-j(+) and J7-h, verify that the K25 EXTERNAL indicator is “LIT”.

4.5.2.2.25 K27 READY TO FIRE indicator. With POWER SUPPLY #2, item 7 of 4.4.5, apply 22.0 ± 0.1 Vdc between J1-e(+) and J7-R, and supplying 22 Vdc, verify that the K27 READY TO FIRE indicator is “NOT LIT”. With POWER SUPPLY #3, item 8 of 4.4.5, apply 22.0 ± 0.1 Vdc between J1-b(+) and J7-R, and supplying 22 Vdc, in addition to the power being supplied by POWER SUPPLY #2, verify that the indicator is "LIT", Disconnect POWER SUPPLY #3 and verify that the indicator remains "LIT" . Disconnect POWER SUPPLY #2 and verify that the indicator is “NOT LIT”.

4.5.2.2.26 SIGHT CAGE indicator. With POWER SUPPLY #2, item 7 of 4*4.5, connected between J6-D (-) and J7-R, and DVM, item 2 of 4.4.5, connected between J6-D(+) and J7-R, and MODE switch set to the RADAR READY position, increase the POWER SUPPLY voltage until the DVM measures -1.25 ± 0.01 Vdc. Verify that the SIGHT CAGE indicator is “NOT LIT”. Increase the POWER SUPPLY voltage until the DVM measures -1.55 ± 0.01 Vdc. Verify that the SIGHT CAGE indicator is “LIT”.

4.5.2.2.27 Torque Motor load. With the IMPEDANCE BRIDGE, item 3 of 4.4.5, connected between J6-K and J6-T, verify 0.12 to 0.18 henry inductance at 400 Hz. With the MULTIMETER connected between J6-K and J6-T, verify 31.5 to 42.5 ohms resistance. With the MULTIMETER connected between J6-K and J6-B, verify 1.9 to 2.1 ohms resistance.

4.5.2.2.28 Magnet load. Connect POWER SUPPLY #1, item 6 of 4.4.5, between J6-M(+) and J6-A and adjust to 20.0 ± 0.1 Vdc. With the DVM, item 2 of 4.4.5, connected between J6-P(+) and J6-D, verify +1.88 ± 0.35 Vdc. With POWER SUPPLY #1 connected in series with the MULTIMETER, item 1 of 4.4.5, between J6-A(+) and J6-M, adjust the POWER SUPPLY to 1.0 ± 0.1 AMPERE D.C. With the DVM connected between J6-A(+) and J6-M, verify +0.9 ± 0.5 Vdc.

4.5.2.2.29 MANUAL RANGE signal. Using POWER SUPPLY #2, item 7, and DVM, item 2, of 4. 4.5; apply the voltage of Table II section

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

3.4.30. With the DVM connected between J7-S(+) and J7-U, set the MANUAL RANGE knob to the settings indicated in 3.4.30. Verify that the DVM indicates the corresponding voltages of 3.4.30 for each MANUAL RANGE knob setting.

4.5.2.2.30 MANUAL RANGE RATE signal. Using POWER SUPPLY #2, item 7, and DVM, item 2, of 4.4.5; apply the voltage of Table II section 3.4.31. With the DVM connected-between J7-T(+) and J7-U, set the MANUAL RANGE RATE knob to the setting indicated in 3.4,31. Verify that the DVM indicates the corresponding voltages of 3.4.31 for each MANUAL RANGE knob setting.

4.5.2.2.31 COSINE signal. Using POWER SUPPLY #2, item 7, RESISTOR item 5, and DVM, item 2 of 4.4.5; apply the voltage and load of Table II, section 3.4.32. With the DVM connected between J4-B(+) and J7-R, set the COSINE knob to the settings indicated in 3.4.32. Verify that the DVM indicates the corresponding voltages of 3.4.32 for each COSINE knob setting.

4.5.2.2.32 LEAD ANGLE TRAVERSE signal. Using POWER SUPPLY #2, item 7, RESISTOR item 5 and DVM, item 2 of 4.4.5; apply the voltage and load of Table II; section 3.4.33. With the DVM connected between J4-L(+) and J4-K, set the LEAD ANGLE TRAVERSE knob to the settings indicated in 3.4.33. Verify that the DVM indicates the corresponding voltages of 3.4.33 for each LEAD ANGLE TRAVERSE knob setting.

4.5.2.2.33 LEAD ANGLE EL signal. Using POWER SUPPLY #2, item 7, POWER SUPPLY #3, item 8, RESISTOR, item 5, and DVM, item 2 of 4.4.5; apply the voltages and load of Table II, section 3.4.3.6, With the LEAD ANGLE EL knob set at “15” use the DVM to verify the voltages of 3.4.34 between J4-V(+) to J4-c. Repeat this procedure to *verify* the voltages of 3.4.34 for LEAD ANGLE EL knob settings of “O” and "-15".

4.5.2.2.34 5.0 Amperes. Set the SYSTEM POWER switch to the, "CFF" position. With POWER SUPPLY #5, item 10 of 4.4.5 connected between J103-F(+) and J10-A, verify that the circuit breaker does not open within a 2 minute period when subjected to a current of 5,75 ± 0.05 amperes D.C. Increase POWER SUPPLY #5 to 10.0 ± 0.1 amperes D.C., verify that the circuit breaker opens within 15 to 55 seconds.

NOTE : This test may be performed prior to the installation of the circuit breaker into the next higher assembly when verification of the test is provided.

4.5.2.2.35 K3 Relay. Connect POWER SUPPLY #1, item 6 of 4.4.5, to pins F (+) and E (-) of connector J103, and adjust for a voltage of +22.0 ± 0.1 Vdc. Using the MULTIMETER, item 1 of 4.4.5, check either resistance or voltage at the various terminals in-

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

dicated in 3.4.36 when the POWER SWITCH and the RANGE/RANGE RATE switch are placed in the indicated positions. Verify the resistance and voltage requirements of 3.4.36.

4.5.3 Temperature extremes. The following tests shall be performed at the test conditions of 4.5.1.1 and 4.5.1.4 as applicable to determine compliance with 3.4.37.

4.5.3.1 LEAD ANGLE GREATER THAN 25° pushbutton switch. With the MODE swith set to the RADAR READY position and the DVM, item 2 of 4.4.5, connected between J4-L(+) and J7-P, verify -8.42 ± 0.50 Vdc with the LEAD ANGLE GREATER THAN 25° pushbutton switch DEPRESSED.

4.5.3.2 RANGE GREATER THAN 3000M pushbutton switch. With POWER SUPPLY #1, item 6 of 4 4.5, apply +28 .0 ± 0.1 Vdc between pins F (+) and E (-) of connector J103. With the DVM, item 2 of 4.4.5, connected between J2-W(+) and J7-P, verify +15.4 ± 1.5 Vdc with the RANGE GREATER THAN 3000M pushbutton switch DEPRESSED. .

4.5.3.3 SIGHT CAGE indicator. With POWER SUPPLY #2, item 7 of 4.4.5, connected between J6-D (-) and J7-R, and DVM, item 2 of 4.4.5, connected between J6-E(+) and J7-R, increase the POWER SUPPLY voltage until the DVM measures -1.10 ± 0.01 Vdc. Verify that the SIGHT CAGE indicator is “NOT LIT”. Again increase the POWER SUPPLY voltage until the DVM measures -1.70 ± 0.01 Vdc. Verify that the SIGHT CAGE indicator is “LIT”.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet 9333052.

6. NOTES

6.1 Intended use. This test set is designed for rackmounting into the AN/TSM-100A Electronics System Test Set, which is used for troubleshooting and performance checking of specific weapon system components.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this and the item detail specifications.

b. Applicable stock number.

c. A block of serial numbers to identify each test set.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

d. Quality Conformance Inspection, if other than specified in Section 4 of item detail specification.

e. First Article Sample requirements, if other than specified in Section 4 of item detail specification.

f. Packaging requirements, if other than specified in Section 5 of-item detail specification.

g" Certificate of conformance for each lot or shipment of product.

6.2.2 Contract data requirements.

a. Contract data requirements for inspection equipment designs conforming to Data Item Description DI-R-1714, tailored, with appropriate addendum to each. See 6.3.

b. Data Cards - When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item Description DI-L-1410.

6.3 Inspection equipment designs. Inspection equipment designs are of two types - Government designs and contractor designs. Government designs are designated by drawing numbers in the appropriate paragraph in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government design is provided; however, unless otherwise specified, the contractor may submit alternate or modified contractor designs in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 Government designs. Government designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in DOD-STD-100.

c. An envelope drawing, as defined *in* DOD-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. Contractor designs are required for all inspection equipment for which Government designs are not

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

specified and include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with Government designs if not otherwise covered-thereby (see 6.3.1c). Designs shall be of the category and form (per DOD-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The item detail specification number, paragraph number and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

6.3.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Center, ATTN: AMSMC-QAF-I (D), Dover, NJ 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as US Army Armament Research and Development Command, (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.5 Changes to previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 4931-A209)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDER
US ARMY ARMAMENT RESEARCH & DEVELOPMENT CENTER
ATTN: DRSAR-QA
DOVER, NJ 07801 5001

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-60878A	TEST SET, SIGHT CURRENT GENERATOR, TS-2660/TSM-100A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:03Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:03Z
Check the source to verify that this is the current version before use.